

22	D	As oil drop rises at constant speed, no net force is acting on the oil drop and drag force acts downward, hence $mg + D = q \frac{V_1}{d}$. As oil drop falls at the same constant speed, no net force is acting on the oil drop, drag force acts upwards and has the same value as before, so $mg = q \frac{V_2}{d} + D$ Adding the two equations, we can eliminate D on both sides, giving. $2mg = \frac{q}{d}(V_1 + V_2)$, so $q = \frac{2mgd}{V_1 + V_2}$
23	D	By potential divider principle, voltmeter reading increase when effective